

Contingency Table (1)

(분할표 (1))

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Lecture Introduction

주별	수업목표	수업내용
1주	강의소개	강의 개요, 기본 개념 소개
2주	분할표 (1)	분할표의 확률구조, 오즈비 (odds ratio)
3주	분할표 (2)	독립성 검정, 정확 검정, 부분 연관성
4주	GLM의 기본 구조	지수족 분포, 선형 예측자, 연결함수
5주	GLM의 모형 진단	모수 추정, 검정, 모형 적합도 평가
	추석 (10/6)	휴강
6주	로지스틱모형 (1)	이항형 반응변수, 로짓함수, 모형 해석
7주	로지스틱모형 (2)	다중 로지스틱 회귀, 교호작용, 예측 확률
8주	Midterm Exam (10/27)	중간고사 (이론)
9주	로그선형모형 (1)	도수형 반응변수, 포아송 회귀
10주	로그선형모형 (2)	다양한 모형 구조 비교 (독립, 조건부 독립, 포화 모형), 모형 선택 기준 및 deviance 분석
11주	로그선형모형 (3)	과산포 (overdispersion), 준가능도 (quasi-likelihood)
12주	혼합효과모형 (1)	반복 측정 자료 개요, 고정효과와 임의효과
13주	혼합효과모형 (2)	선형 혼합 모형
14주	혼합효과모형 (3)	일반화 선형 혼합 모형, 일반화 추정 방정식
15주	Final Exam (12/15)	기말고사 (이론)

중간고사 (30%) - 8주차 / 기말고사 (30%) - 15주차

과제 (30%) - 4번 (3주, 8주, 11주, 14주) / 출석 (10%)

Office Hour: 수 9시-10시 (자연대본관 204호) / kpark@jbnu.ac.kr

Today's Topics

1. Probability Structure for Contingency Table

- Joint, Marginal, and Conditional Probability
- Sensitivity and Specificity
- Independence
- Sampling Schemes
- Research Design for Categorical Data

2. Proportion Structures in a 2x2 Table

- Difference in Proportions, Relative Risk, and Odds Ratio

Probability Structure for Contingency Table

Categorical data consist of observed frequencies in response categories.

A contingency table classifies frequencies across levels of two or more categorical variables.

(Example) When two categorical variables X and Y have I and J levels, respectively

	Y				
	범주	1	...	J	합
X	1	n_{11}	...	n_{1J}	n_{1+}
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	I	n_{I1}	...	n_{IJ}	n_{I+}
	합	n_{+1}	...	n_{+J}	n

Joint and Marginal Probability

Joint distribution

: distribution of two or more variables occurring together

(Example) Joint probability of two categorical variables X and Y

$$\pi_{ij} = P(X = i, Y = j), \quad \sum_{i=1}^I \sum_{j=1}^J \pi_{ij} = 1$$

Marginal distribution

: distribution of a single variable by summing over others

Marginal distribution is obtained by summing the joint probabilities over all other variables except the variable of interest.

(Example) Marginal probability of Y

$$P(Y = j) = \sum_{i=1}^I \pi_{ij} = \pi_{+j}$$

Joint and Marginal Probability (Cont'd)

Joint and Marginal Distributions in Samples

Sample proportions are $p_{ij} = n_{ij}/n$, $p_{i+} = \sum_j p_{ij}$, and $p_{+j} = \sum_i p_{ij}$

	Y				
	범주	1	...	J	합
X	1	p_{11}	...	p_{1J}	p_{1+}
	$\vdots$	$\vdots$	$\ddots$	$\vdots$	$\vdots$
	I	p_{I1}	...	p_{IJ}	p_{I+}
	합	p_{+1}	...	p_{+J}	1

Conditional Probability

Conditional distribution

: distribution describing response (Y) given explanatory variables (X)

$$P(Y = j | X = i) = \frac{P(X = i, Y = j)}{P(X = i)} = \frac{\pi_{ij}}{\pi_{i+}}$$

(Example) Belief in Afterlife

성별	내세의 믿음	
	예	아니오
여성	435	147
남성	375	134

1. Proportion of people who believe in the afterlife
2. Proportion of people who believe in the afterlife among men
3. Proportion of people who are men among believers

Sensitivity and Specificity

Diagnostic tests accuracy

Positive - Test predicts disease

Negative - Test predicts no disease

		Condition (gold standard/Truth)	
		positive (H1)	negative (H0)
Test	Positive (H1)	True positive	False positive
	Negative (H0)	False negative	True negative

Sensitivity and Specificity (Cont'd)

X : disease status ($X = 1$: diseased, $X = 0$: healthy)

Y : test result ($Y = 1$: positive, $Y = 0$: negative)

Sensitivity: $P(Y = 1|X = 1)$ and Specificity: $P(Y = 0|X = 0)$

Positive predictive value: $P(X = 1|Y = 1)$

Negative predictive value: $P(X = 0|Y = 0)$

(Example) A mammogram is used to diagnose breast cancer. The prevalence of breast cancer is 1%, the sensitivity is 86%, and the specificity is 88%. What is the probability that a woman actually has breast cancer given a positive mammogram result?

Independence

Two random variables X and Y are statistically independent if:

$$P(Y=y|X=x) = P(Y=y) \Leftrightarrow P(X=x, Y=y) = P(X=x)P(Y=y)$$

Test of independence

$$H_0: \pi_{ij} = \pi_{i+}\pi_{+j} \quad \forall i,j \quad \text{vs} \quad H_1: \text{not } H_0$$

Test statistics: Pearson chi-square, etc

Test of homogeneity

$$H_0: \pi_{1|j} = \pi_{2|j} = \dots = \pi_{I|j} = \pi_{+j} \quad \text{vs} \quad H_1: \text{not } H_0$$

Test statistics: Pearson chi-square, etc

Remark: The null hypotheses of independence and homogeneity are equivalent
→ The distinction depends on the sampling design

Sampling Schemes for Contingency Tables

Independent Multinomial Sampling

Row totals are fixed in advance

Common in designed experiments (e.g., clinical trials, diet comparison).

(Example) Comparison effects of Diet A and B

	Excellent	Average	Poor	표본크기
Diet A	37	24	19	80(고정)
Diet B	17	33	20	70(고정)
합	54	57	39	150

$P(\text{Diet A, Excellent})$: not estimable

$P(\text{Excellent} \mid \text{Diet A})$: estimable $\rightarrow$ homogeneity test

Sampling Schemes for Contingency Tables (Cont'd)

Multinomial Sampling

Only the total sample size n is fixed.

No restriction on row or column totals.

(Example) Adoption study of 408 children

양부모	양자		합
	오른손잡이	왼손잡이	
모두 오른손잡이	307	48	355
한 명이상 왼손잡이	47	6	53
합	354	54	408(고정)

P (양부모 모두 오른손잡이, 양자 오른손잡이): estimable $\rightarrow$ independence test

Sampling Schemes for Contingency Tables (Cont'd)

Poisson Sampling

Neither row totals nor column totals are fixed

Each cell count follows an independent Poisson distribution

(Example) Traffic accident counts by gender and car color

	운전자		합
	남	여	
흰색	101	48	149
빨간색	247	36	283
합	338	94	432

Conditional on total $n = \sum_{i,j} n_{ij}$, the distribution becomes multinomial

$$X_{11} \sim P(\lambda_{11}), \dots, X_{22} \sim P(\lambda_{22})$$

$$\Rightarrow X_{11}, X_{12}, X_{21} | X_{11} + \dots + X_{22} = n \sim \text{Multi}(n, \pi_{11}, \pi_{12}, \pi_{21})$$

Sampling Schemes for Contingency Tables (Cont'd)

Comparison

Disease	Test		Total
	Positive	Negative	
Present	a	c	$a + c$
Absent	b	d	$b + d$
Total	$a + b$	$c + d$	$a + b + c + d$

Independent Multinomial: $a+c$ and $b+d$ fixed

Multinomial: $a+b+c+d$ fixed

Poisson: a , b , c , and d follow Poisson distribution independently

Sampling Schemes for Contingency Tables (Cont'd)

(Example) We want to study drivers' survival status depending on seat belt usage in car crashes

안전벨트	충돌결과	
	사망	생존
착용		
미착용		

1. How should the data be collected if we analyze using a multinomial sampling model?
2. How should the data be collected if we analyze using an independent binomial sampling model?
3. Compare the multinomial sampling and independent binomial sampling approaches.

Research Designs for Categorical Data

Restrospective Study

Direction: backward in time

Typical design: case–control study

- Divide subjects into cases (disease) and controls (no disease)
- Compare exposure distributions between groups

Cross-Sectional Study

Direction: present time

Collects data on exposure and outcome simultaneously.

Research Designs for Categorical Data (Cont'd)

Prospective Study

Direction: forward in time

Examples

- Prospective cohort study: classify subjects by exposure, follow over time, measure disease incidence.
- Clinical trial: randomly assign treatment vs control, follow outcomes.

Research Designs for Categorical Data (Cont'd)

(Example 1) Classify the following cases into appropriate research designs:

1. In a food poisoning outbreak, company employees who attended a picnic were first checked for food poisoning, and then it was investigated whether they had eaten squid dishes (X).
2. A survey was conducted on transportation methods used by male and female students.
3. To investigate the effect of smoking on lung cancer, a cohort was divided into smokers and nonsmokers, and after a certain period of time (e.g., 10 years), the incidence of lung cancer was recorded

(Example 2) Suppose the proportion of hospital patients admitted with acute myocardial infarction is 5%, and about 70% of these patients are smokers. If 1,000 patients are randomly sampled, on average only about 50 patients with myocardial infarction will be included, among whom roughly 35 will be smokers and 15 nonsmokers. Explain a more efficient study design

Proportion Structures in a 2x2 Table

When comparing two groups with binary outcomes, three main measures are used:

1. Difference in Proportions
2. Relative Risk
3. Odds Ratio

(Example) Among 75 patients who underwent amputation surgery, 40 were treated with phenol (carbolic acid) as a disinfectant and 35 were not. The following data were obtained. Draw inferences about survival rates according to the use of phenol

	생존	사망	합
석탄산 有	34	6	40
석탄산 無	19	16	35

Difference in Proportions

Let $Y_1 \sim B(n_1, \pi_1) \perp Y_2 \sim B(n_2, \pi_2)$,

If sample size are large,

$$p_1 = \frac{Y_1}{n_1} \sim \left(\pi_1, \frac{\pi_1(1 - \pi_1)}{n_1} \right) \perp, p_2 = \frac{Y_2}{n_2} \sim \left(\pi_2, \frac{\pi_2(1 - \pi_2)}{n_2} \right)$$

Difference in Proportions (Cont'd)

Inference for $\pi_1 - \pi_2$

Point Estimate: $p_1 - p_2$

Standard Error Estimate:

$$\hat{\sigma}(p_1 - p_2) = \sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}$$

Confidence Interval (Wald):

$$[p_1 - p_2 - z_{\alpha/2} \hat{\sigma}(p_1 - p_2), p_1 - p_2 + z_{\alpha/2} \hat{\sigma}(p_1 - p_2)]$$

Hypothesis Test: $H_0 : \pi_1 - \pi_2 = 0$

- Wald Test Statistics:
$$\frac{p_1 - p_2}{\sqrt{\frac{p_1(1-p_1)}{n_1} + \frac{p_2(1-p_2)}{n_2}}} \sim N(0,1)$$

- Score Test Statistics:
$$\frac{p_1 - p_2}{\sqrt{p(1-p)\left(\frac{1}{n_1} + \frac{1}{n_2}\right)}} \sim N(0,1) \quad \text{where } p = \frac{Y_1 + Y_2}{n_1 + n_2}$$

Difference in Proportions (Cont'd)

(Example) A random sample of 250 individuals aged 30–40 and another 250 individuals aged 60–70 were surveyed for sleep duration. Using the data below, determine whether the proportion of people who sleep less than 8 hours is higher in the 30–40 age group than in the 60–70 age group, by constructing a confidence interval and performing a hypothesis test.

연령	수면시간		합
	8시간미만	8시간이상	
30–40	172	78	250
60–70	120	130	250
합	292	208	500

Relative Risk (RR)

Useful when probabilities are close to 0 or 1

(Example 1) $\pi_1 = 0.510$ and $\pi_2 = 0.501$

(Example 2) $\pi_1 = 0.01$ and $\pi_2 = 0.001$

Inference for relative risk

Point Estimate: p_1/p_2

Distribution of p_1/p_2 ?

- If $Y_2 = 0$, $p_1/p_2 = \infty$
- For small sample sizes, the distribution tends to be highly skewed

Relative Risk (RR) (Cont'd)

(Example) The results of a comparative study of aspirin and placebo are shown below. Calculate the estimates of the relative risk and the risk difference

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

Odds Ratio (OR)

Odds: $O(\pi) = \frac{\pi}{1 - \pi}$

Properties:

- For $0 < \pi < 1$, $0 < O(\pi) < \infty$
- $O(\pi)$ is an increasing function of π
- If the probability of success is higher, the odds are greater than 1
- Relationship between probability and odds:

$$\pi = \frac{O(\pi)}{1 + O(\pi)}, \quad 1 - \pi = \frac{1}{1 + O(\pi)}$$

(Example) If $\pi = 0.6$, then $O(\pi) = \frac{0.6}{0.4} = 1.5$

→ The probability of success is 1.5 times higher than the probability of failure

Odds Ratio (OR) (Cont'd)

Odds Ratio

In a 2×2 contingency table, the odds ratio is a measure of association between rows and columns

$$OR = \frac{O(\pi_1)}{O(\pi_2)} = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}$$

While the relative risk compares the ratio of two probabilities, the odds ratio compares the ratio of two odds.

Odds Ratio (OR) (Cont'd)

Properties:

For $0 < \pi_1, \pi_2 < 1$, the odds ratio satisfies $0 < OR < \infty$

In a 2×2 contingency table, if the explanatory variable X does not affect the response variable Y , then $\pi_1 = \pi_2 \rightarrow OR = 1$.

$OR = 1$: indicates independence between the two variables.

$OR \gg 1$: $O(\pi_1) \gg O(\pi_2)$, meaning the first category of X has a positive effect (association) on the success of Y relative to the second category.

$OR \ll 1$: $O(\pi_1) \ll O(\pi_2)$, meaning the first category of X has a negative effect (association) on the success of Y relative to the second category.

Odds Ratio (OR) (Cont'd)

Properties:

The odds ratio remains unchanged even if rows and columns are interchanged

In calculating the odds ratio, the two variables are treated symmetrically
→ there is no need to distinguish between explanatory and response variables

Cross-product ratio:

$$\begin{aligned} OR &= \frac{P(Y = 1 | X = 1)/P(Y = 0 | X = 1)}{P(Y = 1 | X = 0)/P(Y = 0 | X = 0)} = \frac{P(Y = 1 | X = 1)P(Y = 0 | X = 0)}{P(Y = 1 | X = 0)P(Y = 0 | X = 1)} \\ &= \frac{P(X = 1 | Y = 1)/P(X = 0 | Y = 1)}{P(X = 1 | Y = 0)/P(X = 0 | Y = 0)} \end{aligned}$$

Remark: In independent multinomial sampling, depending on the data collection scheme, only one of $P(Y = 1 | X = 1)$ or $P(X = 1 | Y = 1)$ can be computed. For example, if data are collected by levels of X , then $P(X = 1 | Y = 1)$ cannot be calculated. However, the estimated odds ratio remains valid and can be interpreted equivalently, regardless of whether sampling was based on X or Y

Odds Ratio (OR) (Cont'd)

Sample Odds Ratio:

$$\hat{\theta} = \frac{p_{11}p_{22}}{p_{12}p_{21}} = \frac{n_{11}n_{22}}{n_{12}n_{21}}$$

Under standard sampling models, $\hat{\theta}$ is the maximum likelihood estimator (MLE) of the true odds ratio OR

(Example) Aspirin vs placebo: Compute the odds ratio and provide an appropriate interpretation

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

Odds Ratio (OR) (Cont'd)

Inference for Odds Ratio and Log Odds Ratio

Distribution of the Odds Ratio

- When the sample size is small, the distribution of $\hat{\theta}$ is highly skewed
→ Inference is usually made on the log odds ratio, $\log \theta$, rather than directly on θ

Properties of the Odds Ratio

For $0 < \pi_1, \pi_2 < 1$, the log odds ratio is symmetric around 0 and approximately normal

If $X_1 \sim B(n_1, \pi_1)$, $X_2 \sim B(n_2, \pi_2)$, then for large samples:

$$\log \hat{\theta} \sim N\left(\log \frac{\pi_1(1-\pi_2)}{\pi_2(1-\pi_1)}, \frac{1}{n_1\pi_1} + \frac{1}{n_1(1-\pi_1)} + \frac{1}{n_2\pi_2} + \frac{1}{n_2(1-\pi_2)}\right)$$

(Example) Compare the cases $\pi_1 = 0.8, \pi_2 = 0.7$ vs. $\pi_1 = 0.7, \pi_2 = 0.8$

Odds Ratio (OR) (Cont'd)

Inference for Odds Ratio and Log Odds Ratio

Confidence Interval for Odds Ratio

- Asymptotic standard error (ASE):

$$\hat{\sigma}_n(\log \hat{\theta}) = \sqrt{\frac{1}{n_{11}} + \frac{1}{n_{12}} + \frac{1}{n_{21}} + \frac{1}{n_{22}}}$$

- For large samples, a $100(1 - \alpha)\%$ confidence interval for is:

$$(L, U) = [\log \hat{\theta} - Z_{1-\alpha/2} \hat{\sigma}_n(\log \hat{\theta}), \log \hat{\theta} + Z_{1-\alpha/2} \hat{\sigma}_n(\log \hat{\theta})]$$

- Corresponding confidence interval for θ :

$$[e^L, e^U]$$

Odds Ratio (OR) (Cont'd)

(Example) For the aspirin and placebo data, construct the confidence interval for the odds ratio

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

Odds Ratio (OR) (Cont'd)

Mean Squared Error (MSE)

A measure used to compare the parameter of interest with its point estimator

$$E\{(\hat{\theta} - \theta_0)^2\} = E\{(\hat{\theta} - E(\hat{\theta}))^2\} + (E(\hat{\theta}) - \theta_0)^2$$

A smaller value indicates a better point estimator.

(Example) Compute the MSE of $\log \hat{\theta}$ when one or more cell counts are zero

When a cell count is zero, the sample odds ratio $\hat{\theta}$ becomes 0, ∞ , or undefined

Corrected odds ratio:

$$\tilde{\theta} = \frac{(n_{11} + 0.5)(n_{22} + 0.5)}{(n_{12} + 0.5)(n_{21} + 0.5)}$$

In the asymptotic standard error formula, replace n_{ij} with $n_{ij} + 0.5$

Odds Ratio (OR) (Cont'd)

(Example) Aspirin vs. placebo: Using the corrected odds ratio, construct the 95% confidence interval for $\log \theta$

약	심근경색		합
	유	무	
위약	189	10,845	11,034
아스피린	104	10,933	11,037

Odds Ratio (OR) (Cont'd)

Relationship Between Odds Ratio and Relative Risk

Similarity between odds ratio and relative risk

$$OR = \frac{\pi_1/(1 - \pi_1)}{\pi_2/(1 - \pi_2)} = \frac{\pi_1(1 - \pi_2)}{\pi_2(1 - \pi_1)}, RR = \frac{\pi_1}{\pi_2}$$

When the success probabilities of both groups are close to 0, the odds ratio and the relative risk take similar values

When the relative risk cannot be directly estimated (e.g., in retrospective studies), the odds ratio can be used as an approximation

(Example) A study compared 262 patients treated for acute myocardial infarction with 519 patients hospitalized for other diseases.

흡연상태	심근경색	대조집단
흡연	172	173
비흡연	90	346
합	262	519

Odds Ratio (OR) (Cont'd)

Remark

X : myocardial infarction status, Y : smoking status

The above result does not directly represent the magnitude of the effect of smoking on myocardial infarction.

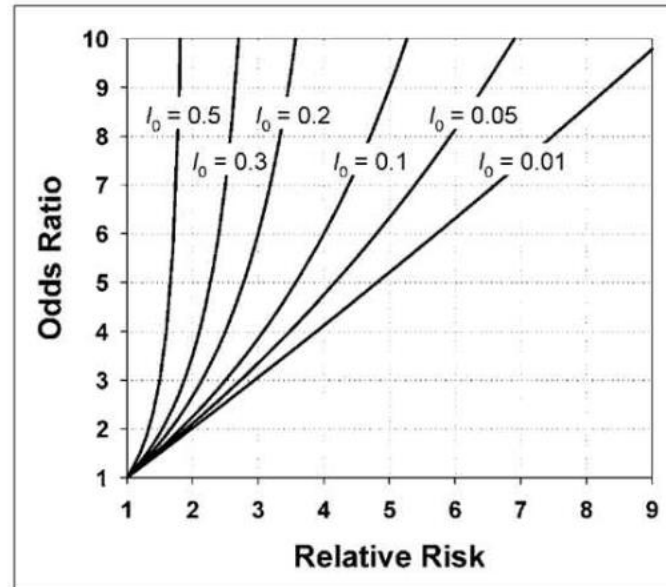
Although $P(X = 1 | Y = 1)$ cannot be estimated in a retrospective study, the following property holds, so the odds ratio from a retrospective study is equal to the odds ratio from a prospective study:

$$\frac{P(Y = 1 | X = 1)/P(Y = 0 | X = 1)}{P(Y = 1 | X = 0)/P(Y = 0 | X = 0)} = \frac{P(X = 1 | Y = 1)/P(X = 0 | Y = 1)}{P(X = 1 | Y = 0)/P(X = 0 | Y = 0)}$$

Moreover, when $P(X = 1 | Y = 1)$ and $P(X = 1 | Y = 0)$ are sufficiently small, the odds ratio from a retrospective study approximately equals the relative risk of smoking

Proportion Structures in a 2x2 Table (Cont'd)

(Example)



I_0 :prevalence, Schmidt and Kohlmann, 2008

When the incidence of myocardial infarction is low, the incidence among smokers can be considered about four times higher than that among nonsmokers